Pass@k Hides Differential Collapse: A Controlled Study of Post-Training in Medical Diagnosis

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Abstract

A clinical diagnosis is a ranked differential, not a single answer, yet language models are mostly judged by whether their top answer is correct. We study what post-training does to the rest of the differential. We compare base models with their post-trained descendants under a controlled design: two base/instruct pairs (Qwen2.5-7B, Qwen3-8B), and three public medical models fine-tuned from the same parent with reinforcement learning, SFT with PPO, or distillation. We evaluate them with 32 samples per case on DDXPlus, whose patients come with a reference differential, and on rare-disease cases from Phenopacket Store. When models answer directly, general post-training raises first-answer accuracy by 17 to 27 points but lowers coverage at larger sample budgets: base models are 10 to 15 points better at pass@32. Asking the post-trained model to reason first recovers this coverage but not the diversity of its samples. The post-trained models' sampled differentials collapse onto one or two diagnoses (answer entropy 0.18 vs. 1.31 for the base at T=1, against 1.87 for the reference), even though their top-1 calibration improves. Medical RL applied after instruction tuning improves pass@k at every k but does not undo the collapse; the same RL applied to the base model avoids it. On rare disease, medical post-training raises stated confidence from 0.51 to 0.66–0.69 while accuracy stays below 2%. Single-answer metrics, including pass@k at small k, therefore rate the post-trained models as strictly better while missing a large loss of diagnostic coverage. In GRPO from Qwen3-8B-Base, a diagnosis reward reaches nearly the official model's top-1 accuracy while keeping more than twice its diversity and beating it by 15–16 points at pass@32. A group-level reward that scores the spread of sampled answers narrows the model less than standard RLVR at equal accuracy.

1 Introduction

Clinicians reason over a differential diagnosis, a ranked set of conditions consistent with the presentation, and the conditions they keep in view determine which tests are ordered and which dangerous possibilities are ruled out. Language-model evaluations in medicine nonetheless score a single answer: multiple-choice accuracy, top-1 diagnosis, or at most pass@k for a small k. Post-training, whether general instruction tuning and RLHF or domain-specific medical reinforcement learning and distillation, is reported to improve these scores [1, 8, 14]. Work on reinforcement learning with verifiable rewards outside medicine suggests a cost: RL often sharpens a model's output distribution, so post-trained models win at pass@1 while their base models solve more problems given many samples [16, 19, 20]. Whether a similar trade-off holds for diagnosis, and whether it shows up in the differential rather than only in pass@k, has not been measured.

We measure it directly. Our design holds the pretrained model fixed and varies only the post-training: (i) two base/instruct pairs, Qwen2.5-7B and Qwen3-8B, which isolate general post-

training; (ii) three public medical models fine-tuned from the same parent, Qwen2.5-7B-Instruct: AlphaMed (rule-based RL), HuatuoGPT-o1 (SFT then PPO), and m1 (SFT distillation), plus AlphaMed's RL applied directly to the base model. We draw 32 samples per case and evaluate each model's sampled differentials against a reference distribution on DDXPlus [7], which ships a differential for every synthetic patient, and against an ontology on rare-disease cases from Phenopacket Store [5].

Findings.

1. **General post-training trades coverage for top-1 accuracy.** In both families the post-trained model is far better on its first answer (+17 to +27 points) but worse at larger sample budgets. Its pass@k curve crosses the base model's at k≈4–8, and at k=32 the base model leads by 10–15 points. All intervals exclude zero (§4.1). The pattern survives a reworded prompt and controls for invalid outputs; prompting for step-by-step reasoning removes the large-k loss but not the loss of diversity (§4.3).

2. **The differential collapses, and single-answer metrics do not show it.** Post-trained models give 1.3–2.6 distinct top answers per 32 samples, against 3.8–8.3 for their base models. Their answer entropy is 0.11–0.49, against 0.78–1.59 for base models and 1.87 for the reference differential. This holds even when sampling at T=1. Their top-1 calibration is *better*, so on every single-answer metric they look strictly superior (§4.2).

3. **Medical RL on an instruction-tuned model adds accuracy without undoing the collapse.** AlphaMed improves on its parent at every k (+3.9 at k=1, +8.4 at k=32), so we see no sharpening tax at the medical stage. Its answer diversity matches its parent's, which is already collapsed. The same RL applied directly to the base model keeps most of the base model's diversity and its can't-miss coverage, so the collapse comes from instruction tuning, not from RL (§4.4).

4. **On rare disease, medical post-training increases overconfidence.** With accuracy below 2% for every model, mean stated confidence rises from 0.51 for the parent to 0.66–0.69 for its RL-trained descendants (§4.5).

5. **A group-level reward avoids the collapse.** Rewarding each GRPO sample for its contribution to the match between the group's sampled answers and the reference differential raises accuracy as much as standard RLVR, while preserving and then widening the differential; per-sample distributional rewards do not (§5).

2 Related work

What RL does to a model's output distribution. Yue et al. [20] find that RL with verifiable rewards improves pass@1 on math and code while base models win at large k. Wu et al. [19] argue that RLVR mostly reweights solutions already in the base model's support. Oh et al. [16] document a "sharpening tax" across 14 base/post-trained pairs on agentic benchmarks. In contrast, Cheng et al. [3] report that RL expands the boundary on domains under-represented in pretraining. In medicine, the only pass@k analysis we know of studies vision-language models and finds that RL mainly sharpens [9]. We give the first text-only medical analysis, and we measure the differential directly rather than only pass@k.

Model	Parent	Post-training	Role
Qwen2.5-7B	n/a	none (pretrained)	base
Qwen2.5-7B-Instruct	Qwen2.5-7B	general SFT + RLHF	instruct; medical parent
AlphaMed-7B [14]	Qwen2.5-7B-Instruct	rule-based RL	medical
HuatuoGPT-o1-7B [1]	Qwen2.5-7B-Instruct	SFT + PPO (verifier)	medical
m1-7B-23K [8]	Qwen2.5-7B-Instruct	SFT distillation	medical
Qwen3-8B-Base	n/a	none (pretrained)	base
Qwen3-8B	Qwen3-8B-Base	general post-training	instruct

Table 1: Models studied. Each pair or family shares its pretrained weights, so differences reflect post-training.

Diversity loss from alignment. Instruction tuning and RLHF reduce output diversity [11, 18]. Related work explains this as typicality bias [21] and documents cross-model homogeneity [10]. We quantify this loss against a clinical reference distribution.

Medical reasoning models and calibration. Medical post-training recipes include verifier-guided SFT and PPO [1], minimalist rule-based RL [14], and test-time-scaling distillation [8]. Reward models and RL tend to favor confident answers [4, 12], and reasoning models are often overconfident [15]. We show that this overconfidence appears in public medical descendants of one parent, on rare disease.

Differential diagnosis benchmarks and distributional RL. DDXPlus [7] was introduced for automatic diagnosis systems trained on differentials [6]. Recent LLM evaluations on it score hierarchical top-k accuracy [13]. Puri et al. [17] train models with GRPO to output several answers per forward pass, rewarding coverage of a gold *set* and set-level calibration, and evaluate on DDXPlus. We instead measure collapse against the reference *distribution* across public post-trained model families. Our group-level reward instead operates on the distribution of independently sampled answers (§5).

3 Experimental setup

Tasks. **DDXPlus** [7]: synthetic patients produced by a rule-based medical knowledge base. Each patient has a true pathology and a reference differential, a probability distribution over the 49 conditions. We use 980 test patients, 20 per condition. Each patient's coded symptoms and history are rendered as a short vignette; we fix known mistranslations in the English evidence descriptions. The prompt lists all 49 conditions and asks for the five most likely, with probabilities, as JSON. **Phenopacket Store** [5] (v0.1.27): published rare-disease cases annotated with HPO phenotypes and an OMIM diagnosis. We use a disease-balanced set of 500 cases covering 418 diseases. Each case is a vignette of present and absent phenotypes, and the model is asked for an open-ended differential. Answers are graded by mapping names to Mondo; we credit an exact match, and separately a "related" match to a specific ancestor group or subtype.

Models. Table 1. All checkpoints are public. For the medical models we verified the parent model from each released configuration.

Pair	Model	pass@1	pass@32	Distinct	Entropy	TV↓	Cov@32	ECE↓
Qwen3-8B, T=1	base	0.282	**0.789**	**6.60**	**1.31**	**0.712**	**0.407**	0.214
	post-trained	**0.556**	0.643	1.53	0.18	0.783	0.236	**0.061**
Qwen2.5-7B, T=1	base	0.239	**0.766**	**8.23**	**1.58**	**0.715**	**0.422**	0.303
	instruct	**0.464**	0.663	2.65	0.49	0.761	0.290	**0.137**
Qwen3-8B, T=0.7	base	0.388	**0.730**	**3.76**	**0.78**	**0.750**	**0.322**	0.121
	post-trained	**0.558**	0.607	1.27	0.11	0.793	0.217	**0.063**

Table 2: DDXPlus, 32 samples per case (980 cases). The reference differential has mean entropy 1.87. Distinct and entropy refer to the top-1 answers across samples; Cov@32 is the share of reference probability mass covered by those answers. ECE is for the stated top-1 probability.

Sampling. We draw n=32 samples per case with vLLM, at T=0.7, top-p 0.95, and at T=1.0, top-p 1.0 (sampling from the model's own distribution). All models use their chat template and the same prompt. Completions are capped at 4,096 tokens (8,192 for m1, whose long reasoning otherwise truncates), and generation stops at the end-of-turn token. We take the last valid JSON list in each completion. Unparsable completions count as incorrect; this affects mainly base models, so base-model pass@k is a lower bound at large k, while post-trained gains at small k are partly format. §4.3 reports format-controlled comparisons.

Metrics. *pass@k*: the unbiased estimator of Chen et al. [2] on the top-1 answer of each sample. *Distinct* and *entropy*: the number of distinct top-1 diagnoses across a case's 32 samples, and the entropy of their empirical distribution, averaged over cases. On DDXPlus, TV: total-variation distance between this empirical distribution and the reference differential. *Coverage@k*: the expected share of the reference differential's probability mass covered by the distinct top-1 answers in k random samples. *Calibration*: expected calibration error (10 bins) of the stated top-1 probability against top-1 correctness. Differences come with 95% paired-bootstrap intervals over cases (2,000 resamples).

4 Results

4.1 General post-training trades coverage for top-1 accuracy

Table 2 and Figure 1 compare each base model with its post-trained counterpart on DDXPlus. In both families the post-trained model is much better on its first answer: +27.5 points for Qwen3 and +22.5 for Qwen2.5 at T=1 (intervals $[+25.3, +29.6]$ and $[+20.7, +24.4]$). The advantage shrinks with k and reverses: the curves cross at k≈4–8, and at k=32 the base model leads by 14.6 points (Qwen3, $[+12.2, +17.1]$) and 10.3 points (Qwen2.5, $[+7.8, +12.8]$). The pattern is the same at T=0.7 for Qwen3, crossing at k=4 with a base lead of 12.2 points at k=32 ($[+9.9, +14.9]$). Unparsable base-model samples count as wrong, so these gaps are lower bounds. Qwen2.5 at T=0.7 behaves the same way: +12.6 points at k=1 $[+10.7, +14.5]$ and a base lead of 9.7 at k=32 $[+7.3, +12.0]$.

4.2 The differential collapses, while top-1 calibration improves

The gap at large k reflects a collapsed differential. Across 32 samples, post-trained models name 1.3–2.6 distinct diagnoses per case, against 3.8–8.3 for their base models. Their answer entropy is 0.11–0.49, against 0.78–1.59 for base models and 1.87 for the reference differential (Table 2).

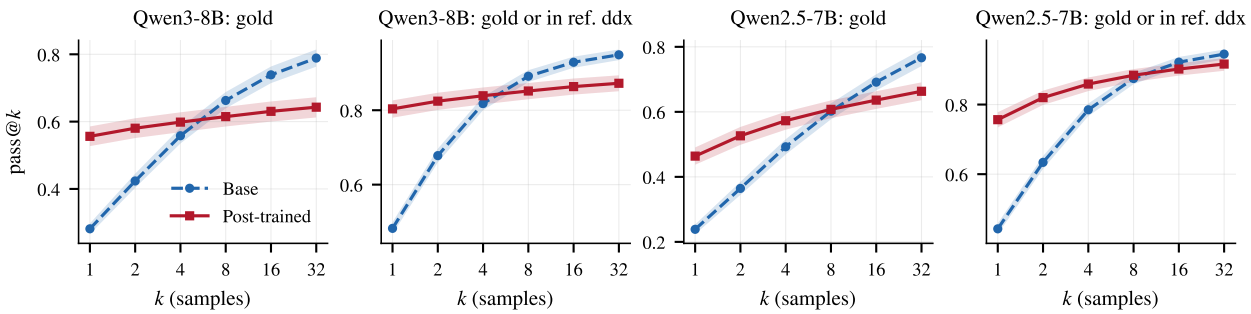


Figure 1: pass@k on DDXPlus at T=1 (980 cases, 32 samples), base model (dashed) vs. its post-trained version (solid), with 95% case-bootstrap bands. "Gold": the top-1 answer is the true pathology; "gold or in ref. ddx": it is the true pathology or appears in the reference differential. In both families the post-trained model wins at small k and loses at large k.

Sampling at T=1 widens the base models' answers but barely changes the post-trained models' (Qwen3 entropy 0.11 at T=0.7, 0.18 at T=1), so the collapse lives in the model's own distribution rather than in the decoding settings. As a result, post-trained models cover less of the reference differential: 0.24 vs. 0.41 of its probability mass at k=32 for Qwen3, and 0.29 vs. 0.42 for Qwen2.5. Their top-1 calibration, however, is markedly better (ECE 0.06 vs. 0.21 and 0.14 vs. 0.30). On every single-answer criterion (accuracy, small-k pass@k, calibration), the post-trained model therefore looks strictly better, and the loss of diagnostic breadth is invisible.

Which diagnoses drop out? For each case and each condition that the reference differential weights at least 0.1, we ask whether the model names that condition as its top answer in at least one of its 32 samples. We call this *considering* the condition. We group the 49 conditions by DDXPlus severity (1 = most severe). Post-training lowers consideration in every tier (Table 3), and the severe "can't-miss" tier is not spared. For the 17 severity-1–2 conditions (e.g. NSTEMI/STEMI, acute pulmonary edema, anaphylaxis, pulmonary embolism, unstable angina), consideration falls from 0.45 to 0.27 for Qwen3 and from 0.52 to 0.34 for Qwen2.5 at T=1. The differences are -18.4 $[-20.8, -15.9]$ and -18.2 $[-21.4, -15.2]$ points. Some diagnoses disappear from the top answer almost entirely. Qwen2.5-7B-Instruct names unstable angina as its top answer in one of 107 eligible cases, where its base model does in 47%. When unstable angina is the true pathology, pass@32 falls from 1.00 to 0.00. Stable angina (0.89 to 0.06) and myocarditis (0.41 to 0.00) follow the same pattern. These conditions are not forgotten: they still appear lower in the listed differential (unstable angina in 50% of eligible cases, against 95% for the base model). The model ranks them but no longer commits to them. At the same time, post-trained models concentrate on a few salient "attractor" diagnoses. Qwen2.5-7B-Instruct gives NSTEMI/STEMI as its top answer in 13.1% of samples, against a reference rate of 4.4% and 6.4% for its base model. It also over-predicts pulmonary embolism (8.2% vs. 3.2%) and pneumonia (8.0% vs. 2.4%). Chest-pain presentations thus collapse onto the most dangerous candidate. That raises its accuracy (NSTEMI pass@1 0.99 vs. 0.44) at the cost of the alternatives, which is a defensible triage bias but not a differential.

4.3 Robustness to prompt and output format

Prompt. We repeat the Qwen3-8B pair with two other prompts (Table 4). *Plain* drops the system prompt and rewords the instructions. *CoT* asks the model to reason step by step before answering; Qwen3's thinking mode stays off in all runs. The plain prompt reproduces the pattern:

Comparison	Considered@32 (base → post-trained) Severe (1–2)	Moderate (3)	Mild (4–5)
Qwen3-8B, T=1	0.45 → 0.27	0.50 → 0.32	0.45 → 0.20
	−18.4 [−20.8, −15.9]	−18.1 [−20.7, −15.7]	−24.2 [−27.0, −21.4]
Qwen2.5-7B, T=1	0.51 → 0.34	0.50 → 0.37	0.38 → 0.23
	−17.0 [−19.7, −14.2]	−12.1 [−14.7, −9.7]	−15.0 [−17.4, −12.6]
Qwen3-8B, T=0.7	0.34 → 0.24	0.44 → 0.29	0.31 → 0.18
	−9.8 [−11.8, −7.9]	−15.8 [−18.2, −13.5]	−13.4 [−15.8, −11.1]
Qwen2.5-7B, T=0.7	0.40 → 0.28	0.41 → 0.32	0.28 → 0.17
	−12.4 [−14.7, −10.2]	−9.1 [−11.2, −6.7]	−10.3 [−12.3, −8.3]
Instruct → AlphaMed, T=0.7	0.28 → 0.32	0.32 → 0.34	0.17 → 0.21
	+4.3 [+2.0, +6.6]	+2.2 [+0.4, +4.0]	+3.9 [+2.4, +5.4]
Instruct → HuatuoGPT-o1, T=0.7	0.28 → 0.36	0.32 → 0.37	0.17 → 0.27
	+8.4 [+6.1, +10.8]	+6.0 [+4.0, +7.9]	+9.5 [+7.5, +11.5]
Instruct → m1, T=0.7	0.28 → 0.41	0.32 → 0.41	0.17 → 0.29
	+13.5 [+10.8, +16.2]	+9.7 [+7.2, +12.2]	+11.4 [+9.2, +13.6]
Base → AlphaMed (base-RL), T=0.7	0.40 → 0.39	0.41 → 0.40	0.28 → 0.27
	−0.7 [−3.0, +1.7]	−0.4 [−2.6, +1.7]	−0.4 [−2.4, +1.7]
Base → `acc` GRPO (8B), T=0.7	0.34 → 0.31	0.44 → 0.42	0.31 → 0.27
	−2.8 [−4.4, −1.3]	−2.1 [−3.7, −0.6]	−4.3 [−6.2, −2.5]
Base → `gm` GRPO (8B), T=0.7	0.34 → 0.32	0.44 → 0.43	0.31 → 0.28
	−1.5 [−3.2, +0.1]	−1.3 [−3.0, +0.3]	−3.0 [−4.8, −1.2]

Table 3: Share of (case, condition) pairs with reference probability ≥ 0.1 in which the condition is a top-1 answer in at least one of 32 samples, by DDXPlus severity. Second rows give differences in points with 95% case-bootstrap intervals. All rows use the 980 test cases; GRPO rows are seed 0. General post-training lowers consideration in every tier, including the can't-miss tier; medical post-training raises it; our GRPO from the base model loses at most 4 points, and with the group reward the severe-tier loss is not significant.

the post-trained model gains +23.0 [+20.9, +25.1] points at k=1 and loses −7.1 [−9.3, −5.0] at k=32 (T=0.7; at T=1, +34.2 and −8.7 [−11.0, −6.4]), with 1.3–1.5 distinct answers against 3.7–5.6. Asking for reasoning changes the picture. With CoT, the post-trained model's pass@32 rises from 0.607 to 0.835: it leads the base model at every k at T=0.7 (+4.7 [+2.9, +6.6] at k=32) and ties it at T=1 (0.0 [−2.1, +1.9]). Its samples remain far less diverse than the base model's (2.1–2.2 vs. 3.9–6.3 distinct answers), so reasoning recovers coverage at k=32 without restoring sampling diversity. The coverage loss is therefore largest when the model answers directly, as in most clinical decision-support deployments, and reasoning at inference time mitigates it. This matches the medical models: the reasoning-trained descendants widen the differential (§4.4).

Output format. Base models more often give invalid answers: unparsable, or with a top answer outside the 49 conditions. This affects 12% of Qwen3-8B-Base samples at T=0.7 and 25% at T=1 (Qwen2.5-7B: 1% and 21%), against at most 0.3% for the post-trained models. Invalid samples count as wrong, which deflates the base model at large k but inflates the post-trained model's gain at small k. Among valid samples only, the post-trained models still gain +12.0 [+9.8, +14.1] (Qwen3, T=0.7), +18.7 [+16.6, +20.7] (Qwen3, T=1), +12.2 [+10.4, +14.0] and +16.8 [+15.0, +18.5] points (Qwen2.5) at k=1, so a quarter to a third of the raw gain at T=1 is format. On the cases where every sample of both models is valid, the large-k loss persists: −15.2 [−21.9, −8.6] at k=32 for

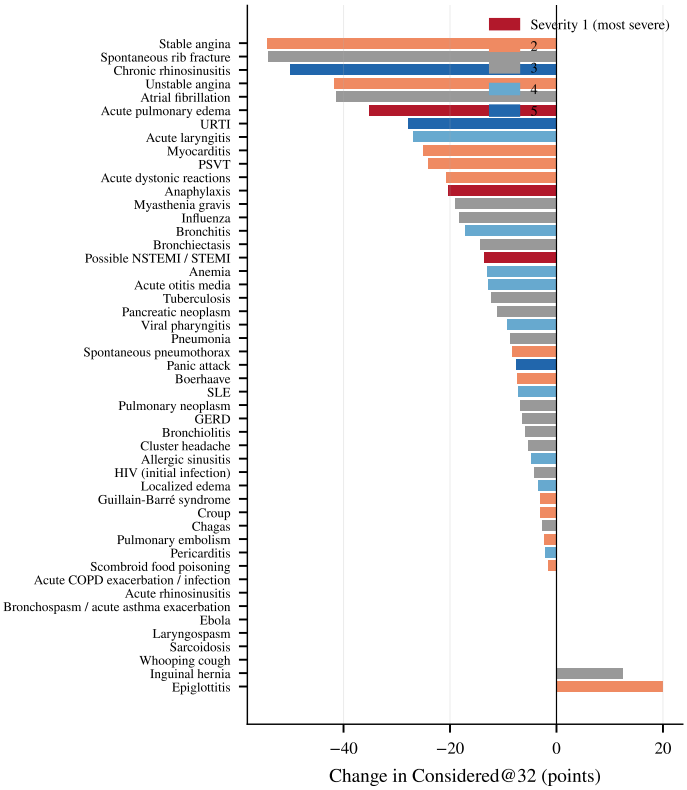


Figure 2: Change in Considered@32 per condition, Qwen2.5-7B-Instruct minus its base model (T=1, 980 cases), colored by DDXPlus severity (dark red = most severe). Conditions with fewer than five cases giving them reference probability ≥ 0.1 are omitted. Angina, acute pulmonary edema, anaphylaxis and myocarditis are among the largest drops.

Qwen3 (105 cases) and -10.2 $[-12.7, -7.6]$ for Qwen2.5 (748 cases), both at T=0.7. (At T=1 almost no case has 32 valid base samples.) The reported large-k losses are thus lower bounds.

4.4 Medical post-training

Applied after instruction tuning, medical RL does not cost coverage. AlphaMed improves on its parent at every k (Figure 3): +3.9 points at k=1 ($[+2.2, +5.5]$), widening to +8.4 at k=32 ($[+6.0, +10.6]$). It reaches at least one correct sample for 47 of 49 conditions, against 39 for its parent. It does not restore breadth, though: distinct answers (1.87 vs. 1.82), entropy (0.29 vs. 0.30), and coverage (0.27 vs. 0.24) stay at the instruction-tuned level, far below the base model. AlphaMed also names off-list conditions more often than its parent, a mild loss of instruction following. By condition, AlphaMed slightly *raises* consideration in every severity tier (+2.2 to +4.3 points; Table 3). It still loses acute pulmonary edema (0.31 to 0.13), and it introduces an attractor of its own: spontaneous pneumothorax as the top answer in 10.3% of samples, against 4.3% for its parent and a reference rate of 1.0%. Medical post-training thus operates on an already collapsed distribution; the collapse comes from the general post-training stage. The other two descendants go further (Table 5). HuatuoGPT-o1 (SFT then PPO) and m1 (SFT distillation from a reasoning teacher) both *widen* the differential relative to their parent: 2.49 and 3.29 distinct answers vs. 1.82, and coverage@32 of 0.31 and 0.33 vs. 0.24. Their pass@32 rises by 13.7 $[+11.3, +16.0]$ and 18.1 $[+15.5, +20.7]$ points. m1 gains almost nothing at k=1 (+0.9, interval includes zero), so its

		pass@1		pass@32		Distinct	
Prompt	T	Base	Post	Base	Post	Base	Post
default	0.7	0.388	**0.559**	**0.730**	0.607	3.76	1.27
default	1.0	0.282	**0.556**	**0.789**	0.643	6.60	1.53
plain	0.7	0.352	**0.583**	**0.727**	0.655	3.72	1.28
plain	1.0	0.244	**0.586**	**0.792**	0.705	5.57	1.54
CoT	0.7	0.409	**0.652**	0.788	**0.835**	3.89	2.08
CoT	1.0	0.275	**0.642**	0.837	0.837	6.31	2.24

Table 4: Qwen3-8B-Base vs. Qwen3-8B on DDXPlus (980 cases, 32 samples) under three prompts. Bold marks the significantly better model (paired case bootstrap). Without reasoning, the base model wins at k=32; with CoT, the post-trained model matches or beats it, but its samples stay far less diverse.

Model	Post-training	pass@1	pass@32	Distinct	Entropy	Cov@32	ECE
Qwen2.5-7B-Instruct	(parent)	0.478	0.598	1.82	0.30	0.239	**0.106**
AlphaMed-7B	rule-based RL	0.516	0.682	1.87	0.28	0.266	0.121
HuatuoGPT-o1-7B	SFT + PPO	**0.536**	0.735	2.49	0.44	0.306	0.168
m1-7B-23K	SFT distillation	0.486	0.779	3.29	0.62	0.328	0.206
Qwen2.5-7B (base)	(parent)	0.352	0.695	**4.91**	**1.00**	**0.335**	0.291
AlphaMed-7B (base-RL)	rule-based RL	0.495	**0.790**	3.14	0.62	0.320	0.202

Table 5: Medical models on DDXPlus (980 cases, 32 samples, T=0.7). Upper block: descendants of Qwen2.5-7B-Instruct; none narrows the differential further, and the reasoning-trained models widen it. Lower block: AlphaMed's RL applied directly to the base model, which keeps most of the base model's diversity and has the best pass@32 of any 7B model.

improvement lies entirely in breadth. Both reasoning-trained models are also less calibrated (ECE 0.17 and 0.21 vs. 0.11). Medical post-training on an instruction-tuned model thus never adds to the collapse, and reasoning distillation partially reverses it.

Medical RL from the base model. AlphaMed was also released as a model trained with its rule-based RL directly from Qwen2.5-7B, skipping instruction tuning [14]. [TODO: confirm in the AlphaMed paper that the base and instruct variants share data and hyperparameters] This gives a third-party test of whether RL itself causes the collapse (Table 5, lower block). It does not. RL from the base model improves on the base at every k ($+14.3$ $[+13.1, +15.7]$ points at k=1, $+9.5$ $[+7.4, +11.6]$ at k=32). It keeps 3.14 distinct answers, against 1.87 for the same recipe applied to the instruct model. Against that instruct-model AlphaMed, it is 2.2 $[0.4, 3.9]$ points worse at k=1 and 10.8 $[+8.6, +13.2]$ points better at k=32. It also preserves can't-miss consideration: severe-tier Considered@32 changes by -0.7 $[-3.0, +1.7]$ points relative to the base, against -12.4 for instruction tuning and -8.1 for instruction tuning followed by AlphaMed's RL (Table 3). This independently reproduces our own GRPO-from-base result (§5): with a verifiable diagnosis reward, RL from a base model buys accuracy without the collapse, which is introduced by general instruction tuning and inherited by everything trained on top of it.

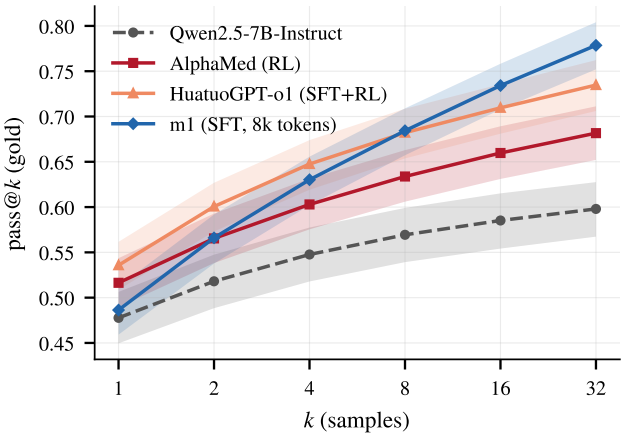


Figure 3: DDXPlus at T=0.7 (980 cases): Qwen2.5-7B-Instruct (dashed) vs. its medical descendants, with 95% case-bootstrap bands. Every descendant matches or beats its parent at every k, and the gaps widen with k: there is no sharpening tax at the medical stage.

4.5 Rare disease: overconfidence without accuracy

On 500 Phenopacket Store cases covering 418 diseases, every 7–8B model is near the floor: pass@1 is 1.3–2.2% and pass@32 is 4–8%. Stated confidence is not near the floor. Qwen2.5-7B-Instruct states a mean top-1 probability of 0.51, and its RL-trained medical descendants state 0.66 (HuatuoGPT-o1) and 0.69 (AlphaMed), while their accuracy stays at 1.9%. ECE accordingly rises from 0.50 to 0.64–0.67. Medical RL thus increases overconfidence exactly where the model is out of its depth. The effect does not depend on the starting point: AlphaMed trained from the base model is the most overconfident model we evaluate (mean confidence 0.71, ECE 0.70, accuracy 1.4%). The one descendant trained only with supervised reasoning traces, m1 (with its intended 8k-token budget), behaves differently. It is the only model that beats the parent at every k (+0.6 [+0.2, +1.1] points at k=1, +2.0 [+0.2, +3.8] at k=32), it is the most diverse (15.4 distinct answers per 32 samples, against 11.5 for the parent and 9.5 for HuatuoGPT-o1), and its confidence rises less (0.56).

General post-training shows a weak version of the DDXPlus pattern. Qwen3-8B beats its base at pass@1 (+0.7 [+0.2, +1.3]) and trails it at pass@32 (−1.4 [−3.0, +0.2]), with half the distinct answers (6.2 vs. 13.5); for Qwen2.5 no pass@k difference is significant. Unlike medical RL, general post-training *lowers* stated confidence here (0.61 to 0.46 for Qwen3, 0.60 to 0.51 for Qwen2.5), although base models also fail the output format more often (8.5% of Qwen3-8B-Base samples). Our 8B GRPO runs (§5) are indistinguishable from their base on this set (all pass@k differences within ±0.6 points).

Accuracy depends strongly on when the source case report was published, which suggests memorization. Across the Qwen2.5 family, pass@32 is 8.1–10.8% on cases published up to 2022 (n=334), 0.9–2.6% on 2023–24 cases (n=115), and 3.9% on 2025+ cases (n=51). Even the low absolute accuracy is therefore inflated by older, likely-memorized cases. [TODO: full 1,401-case set]

5 Training against the collapse

Can reinforcement learning raise diagnostic accuracy without narrowing the differential? We fine-tune Qwen3-1.7B with GRPO on DDXPlus training patients (9,800 cases, 200 per condition; disjoint from the test split) and evaluate exactly as in §3.

Setup. We train LoRA adapters (r=32) with GRPO for 200 steps. Each step uses 16 prompts and 16 sampled completions per prompt, generated with vLLM on the same GPU. A first sweep with a common aggressive configuration (learning rate 2×10^{-5}, no KL penalty, 8 samples per prompt, per-group reward standardization) was unstable. In three of seven runs, the policy collapsed to an input-independent answer: one condition for 88–100% of all patients, with completions shrinking to about 25 tokens. The starting policy is nearly deterministic (token entropy ≈0.06), so most GRPO groups have zero reward variance, and noisy updates drive the collapse. We therefore use learning rate 5×10^{-6}, KL coefficient 0.04, no reward standardization (as in Puri et al. [17]), and 16 samples per prompt. A monitor tracks the number of distinct top-1 answers per case within each batch. All runs in this configuration were stable.

Rewards. Every recipe includes a format reward: the answer must parse into a list of distinct conditions with valid probabilities. `acc` is standard RLVR: 1 if the top-1 answer is the true pathology. `dist` adds two per-sample distributional terms to `acc`: a Brier score of the stated distribution against the truth, and the overlap of the stated distribution with the reference differential. Both are computed from a *single* completion. They cannot see that the policy gives the same answer in every sample, which is exactly the collapse of §4.2. `gm` adds to `acc` a *group-level* reward over the GRPO group of n samples for a case. If sample i's top-1 answer is a, and k_a samples in the group give a,

$$r_i^{\mathrm{gm}} = \mathrm{clip}\big(n\, p_{\mathrm{ref}}(a) - (k_a - 1),\ 0,\ 1\big), \tag{1}$$

where p_{ref} is the reference differential. This is sample i's marginal contribution to $n\sum_c \min(q_c, p_{\mathrm{ref}}(c))$, the overlap between the group's empirical answer distribution q and the reference. An answer earns credit while its condition is under-represented relative to its reference mass, and none once it is over-represented. In contrast to multi-answer rewards that score a set inside one completion [17], `gm` acts on the distribution of *sampled* answers, the quantity that collapses. Because the group reward fires only when a group contains dissenting samples, we also test a boosted variant, `gm`$^+$: `gm` weight 3 and training temperature 1.2. As baselines we adapt the multi-answer rewards of Puri et al. [17] to our prompt. Multi-RLVR scores the share of listed conditions that belong to the reference differential, treated as a set of acceptable answers; Multi-RLCR adds their set-level Brier term on the per-answer probabilities. Both zero the reward for duplicate or invalid answers, as in the original, and use the same format reward, optimizer and settings as our recipes.

Results. Table 7 reports means over two seeds. All recipes raise top-1 accuracy by 8–10 points. The per-sample recipes (`acc`, `dist`) slightly *reduce* diversity relative to the starting model, the sharpening documented in §4.1. The group reward removes this. `gm` matches the starting model's diversity, and `gm`$^+$ exceeds it while also being the most accurate recipe. Against `acc`, with paired bootstraps over cases averaged over seeds, `gm`$^+$ improves:

- pass@1 by +1.6 [+1.0, +2.3] and pass@32 by +1.7 [+1.0, +2.6] points;
- distinct answers by +0.10 [+0.07, +0.13] and entropy by +0.034 [+0.022, +0.047];
- coverage@32 by +1.5 [+1.0, +2.1] points, and TV by −0.014 [−0.019, −0.010].

Plain `gm` improves on `acc` in the same direction with smaller margins (e.g. pass@1 +1.0 [+0.6, +1.6], entropy +0.013 [+0.005, +0.021]). Diversity increases monotonically with group-reward pressure, with no accuracy cost. The set-membership baselines of Puri et al. [17] train stably (their reward rises from 0.27–0.31 to 0.32–0.35) but barely change sampled behavior. pass@1 rises by under 1 point over the untrained model, and diversity is unchanged. Against them, `gm`$^+$ is better on every

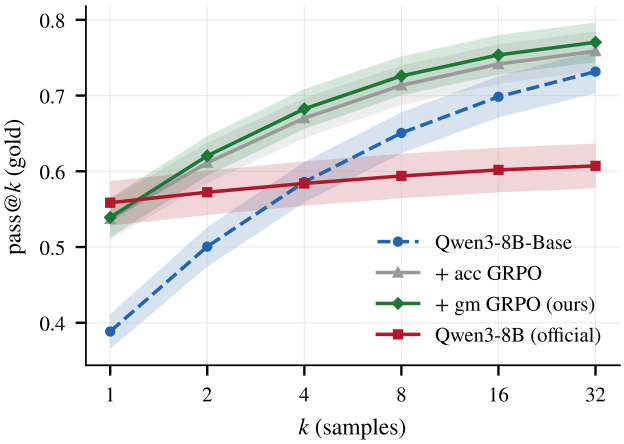


Figure 4: pass@k on DDXPlus (T=0.7, 980 cases) from the same pretrained weights: Qwen3-8B-Base, our GRPO runs (pooled over two seeds), and the official post-trained Qwen3-8B, with 95% case-bootstrap bands. Our RL lifts the whole curve; official post-training trades the right end for the left.

metric, by +8.9 [+7.4, +10.5] points at pass@1, +0.07 [+0.03, +0.11] distinct answers and +3.0 [+2.3, +3.8] points of coverage@32 (vs. MULTI-RLVR; MULTI-RLCR is similar). What they do improve is the differential written *inside* one completion: the reference mass covered by a single listed differential rises to 0.317 (MULTI-RLVR), against 0.300 for `acc` and 0.309 for `gm`$^+$. The two objectives are complementary: a set reward shapes the list a model writes, while the group reward shapes which answer it commits to across samples, which is what collapses under post-training. The absolute effects are modest because the starting policy is already collapsed: there is little exploration for any reward to act on. None of the recipes changes rare-disease accuracy or confidence on Phenopacket Store, where a 1.7B model is at the floor; the same holds at 8B (§4.5).

Starting from a diverse base model. The 1.7B experiment starts from an already collapsed instruct model. We therefore repeat the comparison from Qwen3-8B-Base, the diverse base model of §4.1, with the same recipe and `acc` vs. `gm`. We compare against Qwen's own post-trained Qwen3-8B (Table 6; means of two seeds). Both RL recipes reach nearly the official model's top-1 accuracy (0.537 and 0.539 vs. 0.558; `gm`: −1.9 [−3.9, +0.1] points). They keep more than twice its diversity (2.7–2.9 vs. 1.27 distinct answers) and beat it by 15–16 points at pass@32 (`gm`: +16.3 [+14.0, +18.8]). The collapse produced by general post-training is therefore not the price of its top-1 accuracy. Both runs still narrow the base model somewhat, and the group reward narrows it less. At equal pass@1 (+0.2, n.s.), `gm` beats `acc` on distinct answers +0.14 [+0.11, +0.17], entropy +0.035 [+0.029, +0.041], coverage@32 +1.1 [+0.8, +1.5] points, and pass@32 +1.2 [+0.5, +1.9]. Each seed reproduces the ordering. The severe tier tells the same story (Table 3): official post-training removes 9.8 points of can't-miss consideration, `acc` 2.8, and `gm` 1.5 (not significant). Figure 4 shows the pass@k curves.

6 Limitations

DDXPlus is synthetic: its reference differential is the posterior of a rule-based engine, and the simulated pathology is not always ranked first in it (it is for "more than 70%" of patients [7, §3.5]), so we treat TV and coverage as distance to a *reference* posterior rather than to clinical truth. Base

Model	pass@1	pass@32	Distinct	Entropy	Cov@32	TV↓
Qwen3-8B-Base (start)	0.389	0.732	**3.77**	**0.78**	**0.323**	0.749
+ `acc` GRPO (ours)	0.537	0.759	2.72	0.52	0.297	0.750
+ `gm` GRPO (ours)	0.539	**0.770**	2.86	0.56	0.308	**0.742**
Qwen3-8B (official post-training)	**0.558**	0.607	1.27	0.11	0.217	0.793

Table 6: GRPO from Qwen3-8B-Base vs. the official post-trained Qwen3-8B on DDXPlus (980 cases, 32 samples, T=0.7); GRPO rows are means of two seeds. Our RL nearly matches the official model's top-1 accuracy while keeping more than twice its diversity; the group reward preserves more than `acc`.

Recipe	pass@1	pass@32	Distinct	Entropy	Cov@32	TV↓
Qwen3-1.7B (untrained)	0.378	0.435	1.41	0.151	0.162	0.850
`acc`	0.458	0.499	1.36	0.142	0.181	0.829
`dist`	0.458	0.502	1.34	0.131	0.181	0.829
MULTI-RLVR [17]	0.385	0.443	1.39	0.153	0.166	0.845
MULTI-RLCR [17]	0.383	0.441	1.39	0.150	0.165	0.847
`gm`	0.469	0.510	1.40	0.155	0.186	0.825
`gm`$^+$	**0.474**	**0.516**	**1.46**	**0.176**	**0.196**	**0.815**

Table 7: GRPO on Qwen3-1.7B, evaluated on DDXPlus test (980 cases, 32 samples, T=0.7); means of two seeds. `gm`$^+$: group-reward weight 3, training temperature 1.2. Per-sample rewards slightly narrow the model; set-membership rewards barely move it; the group reward preserves, then widens, its differential while improving accuracy.

models more often give invalid answers. We count these as errors, which makes the large-k losses conservative but inflates post-trained gains at small k; conditioning on valid samples removes a quarter to a third of the T=1 pass@1 gain (§4.3). The base/instruct pairs are Qwen-family 7–8B models; prompt robustness is tested on one pair, and the collapse at large k largely disappears when the model is asked to reason first (§4.3). [TODO: Llama-3.1-8B base/instruct pair (pending access); Llama medical descendants are run.] Phenopacket cases are drawn from published literature and may be in pretraining data, which we partly address with era splits. We do not evaluate clinical utility with clinicians.

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